

Auntor Chakma

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SUMMARY

I work on trustworthy multimodal AI. My recent papers cover federated learning and explainable AI for healthcare, and I am moving toward vision-language models and AI safety. I grew up in Rangamati, in the Chittagong Hill Tracts, where specialist care is hard to reach, so I care most about models that a health worker in a low-resource clinic could run, understand, and trust. I am seeking a PhD position in this area, after a degree delayed by about 3 years by COVID closures and session backlogs.

EDUCATION

Rangamati Science and Technology University (RMSTU) B.Sc. (Engg.) in Computer Science and Engineering; CGPA 3.50/4.00 <i>Four-year program extended by COVID closures and session backlog; result published in 2025</i>	Rangamati, Bangladesh 2018 – 2025
Notre Dame College , Dhaka, Higher Secondary Certificate (HSC), GPA 5.00/5.00	2017
Laker's Public School and College , Dhaka, Secondary School Certificate (SSC), GPA 5.00/5.00	2015

PUBLICATIONS

- Auntor Chakma**, F. T. Z. Moon, F. Jahan, M. K. Morol, T. H. Liew, D. Nandi, A. A. Jubair, M. Rahman. “Multimodal Federated Learning with Privacy-Preserving Mechanisms in Healthcare: A Mini Review of Convergence, Challenges, and Future Directions.” *Frontiers in Digital Health*. **Accepted, Sep 2026**, as first author. Review of 14 studies from seven databases. Federated training held steady at real scale (one federation across 71 hospitals) and multimodal models beat single-modality ones in most direct comparisons, yet only 1 of 8 empirical studies (12.5%) applied a formal privacy mechanism to model updates, and none gave different data types their own protection, so a scan and a heart-rate reading were treated the same.
- Auntor Chakma**, D. Rahman, T. H. Liew. “An Ensemble Approach for Diabetic Retinopathy Severity Classification with Diffusion-Based Augmentation and Explainability.” *HighTech and Innovation Journal* (Q1, Scopus-indexed). **Accepted, Aug 2026**, as first author. Public fundus data is mostly normal and mild cases, so models miss advanced stages. The paper generates rare-stage images with Stable Diffusion 2.1 and ControlNet, teaches that severity progresses, directs attention to lesions, and stacks three backbones. On 1,437 fundus photographs it reaches 91.63% accuracy and AUC 0.9920 in seven-stage classification, cuts the ordinal model's mean error from 0.78 to 0.46 stages, and beats standard augmentation by 4.65% absolute accuracy at matched training size.
- S. Mohajan, G. M. S. Hossain, M. Mynoddin, R. Chakma, T. Mahmud, P. Kanungoe, **Auntor Chakma**. “A Deep Learning-Based Framework for Bangla Vehicle License Plate Detection and Recognition.” ICDSAIA 2025, *Communications in Computer and Information Science*, vol. 2683, Springer, pp. 220–235. **Published, Jan 2026**. Image restoration plus sequence recognition reach 96% character-level accuracy on a synthetic Bangla plate dataset. Role in model development, writing, and figures. [\[Paper Link\]](#)
- Auntor Chakma**. “Attentive TabNet Framework for Multiclass Intrusion Detection in the Internet of Medical Things.” **B.Sc. Thesis**, Rangamati Science and Technology University. TabNet with GAN-generated minority samples reaches 97.96% accuracy on 89 million records.

HONORS, SCHOLARSHIPS AND AWARDS

Chittagong Hill Tracts Development Board Scholarship (academic excellence)	2018, 2022, 2023, 2024
Rangamati Hill District Council Scholarship (academic performance)	2023, 2025
Rangamati Science and Technology University Merit Scholarship	2023, 2024

RESEARCH EXPERIENCE

ELITE Research Lab LLC (explainable and interpretable AI research lab) <i>Researcher & Research Assistant</i> o Extending the research into multimodal AI, centred on vision-language models that show what an answer relies on, expose where they fail, and protect the data they learn from.	Queens, NY, USA (Remote) Apr 2026 – Present
<i>Junior Research Assistant</i> o Led a project under the supervision of Md Kishor Morol, still in progress. Hospitals want one shared model without exchanging patient records, but shared updates can still leak information, so the project adds differential privacy and explanation methods.	Mar 2026 – Apr 2026
<i>AI Research Intern</i> o Examined 14 studies on multimodal federated learning across seven databases and identified the open problem that now drives the research. Learned to run a systematic review and write a review paper.	Jan 2026 – Mar 2026
<i>Research Collaborator</i> o Worked on why retinopathy models miss advanced stages when public data is mostly normal and mild cases. Learned to split a failure into separate problems, test each fix by removing it, and handle heavily imbalanced imaging data.	Aug 2025 – Dec 2025
Rangamati Science and Technology University <i>Research Assistant, Dept. of CSE</i> o Built a real-time network traffic dataset from Bangladesh telecommunications and trained CNN intrusion detectors. Learned the pipeline from raw data to evaluation on messy, imbalanced traffic, and identified research gaps through the literature review.	Rangamati, Bangladesh Dec 2024 – Jun 2025

SKILLS

- **Programming** Python, Java, C++
- **Methods** Federated learning, differential privacy, explainable AI, vision-language models, ordinal regression, diffusion-based augmentation
- **ML/DL** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face Transformers, LangChain, LangGraph, RAG
- **Data and Tools** NumPy, Pandas, Matplotlib, Seaborn, Linux, Git, LaTeX, Wireshark

REFERENCES

- **Md Kishor Morol**, CEO, ELITE Research Lab LLC; Assistant Professor (on leave), American International University-Bangladesh. kishor@elitelab.ai
- **Dr. Tanjim Mahmud**, Associate Professor, Dept. of CSE, RMSTU. tanjim.cse@rmstu.ac.bd
- **Md Mynoddin**, Assistant Professor, Dept. of CSE, RMSTU. mynoddin@rmstu.ac.bd